

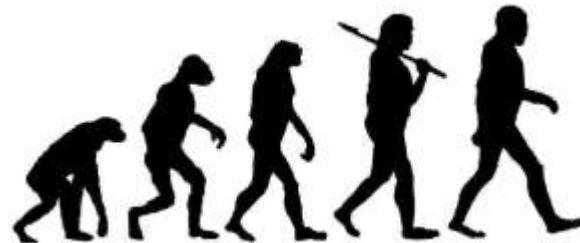
CSE 331

AUTOMATA AND COMPUTABILITY

Lecture 1

Spring 2026

Learning goals



Introductions



Logistics

Homework: 2 (one HW before mid, one HW after mid)

In-class: Group discussion + **polls** ; Read **book** first

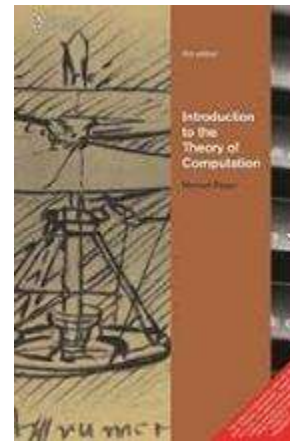
Discussion section: more examples, homework warmup

Exams: Midterm, **Apr 2, 2026 8:30 AM - 10:30 AM**
Final Exam, **May 16, 2026 8:30 AM - 10:30 AM**

Google Classroom: Homework submission

Google Classroom : Announcements and Q&A.

Consultation: with me (check consultation routine).



Grading

- Your final grade has 4 components
- Final exam: 40%
- Midterms: 30%
- Quiz: 20%
- Homework: 10% (Both of the HW will be counted)

How to excel

- Prepare ahead of class
 - Read assigned sections, read homework questions
- Engage in class
 - **Discuss questions with your neighbors (when we are in person),** look for (counter) examples
 - Go over wrong choices too!
- Reinforce after class
 - Briefly summarize what you learned
- Start homework early
 - Tackle problems together: brainstorm, plan, and solve together
- Seek help and seek to help others, with integrity



Learning
How to Learn

How to excel with integrity

It's an integrity violation to...

- Ask others to give you specific HW or test answers
- Share your answers on HW or test
- Work on HW with anyone other than your HW partners
- Search the internet or other resources not provided for the class for HW solutions
- Share answers or notes while taking an exam

This not a complete list ... you are responsible for knowing and following the guidelines *Academic integrity violations will be taken seriously and reported immediately*

Homework

- Homework is 10% of your grade
- The main goal of homework is to help you learn the material in your own pace
- Exams are going to be VERY SIMILAR to homework problems, so it is better for you to think on these problems when you have time, and not during the exam

Homework and AI

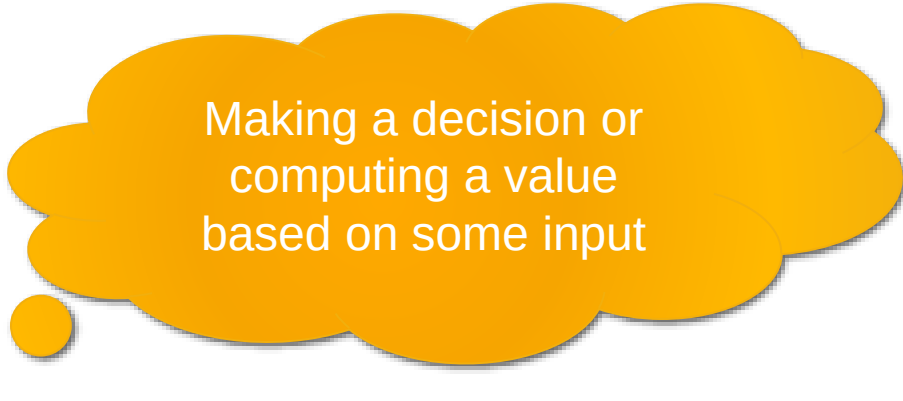
- It is forbidden to use ChatGPT or similar tools for homework
- If we catch students who use these, or cheat in any other way, they will be reported to the academic integrity review board

Participation

- We need to maintain 70% of the attendance in class according to university policy to sit for final exam
- In class we will have questions and follow-up discussions which will help you understand concepts better

CSE 331's big questions

- What problems are computers capable of solving?
- What resources are needed to solve a problem?
- Are some problems harder than others?



Making a decision or
computing a value
based on some input

Questions about algorithms

Are they correct (for given specification)?

Is there a better approach to solving the same problem?

Does each problem have a solution?

If so, can that solution be found by an algorithm?

Questions about algorithms

Are they correct (for given specification)?

CSE230: Discrete Math

Is there a better approach to solving the same problem?

CSE 221: Algorithms, **Complexity theory**

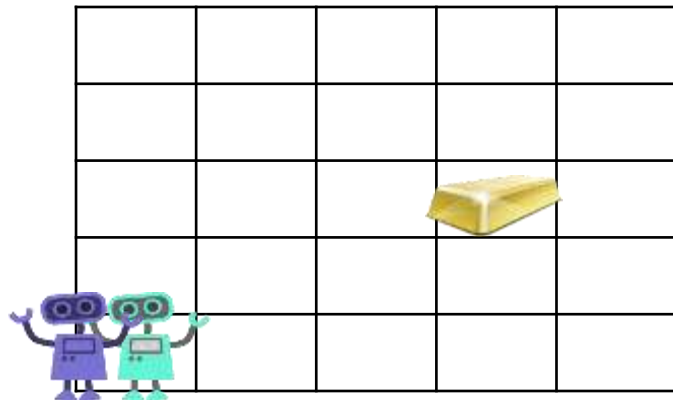
Does each problem have a solution?

CSE 331: Automata and Computability

If so, can that solution be found by an algorithm?

CSE 331: Automata and Computability

Activity



Goal: find treasure hidden somewhere in grid. Describe an algorithm for each robot.

Do you need to assume anything?

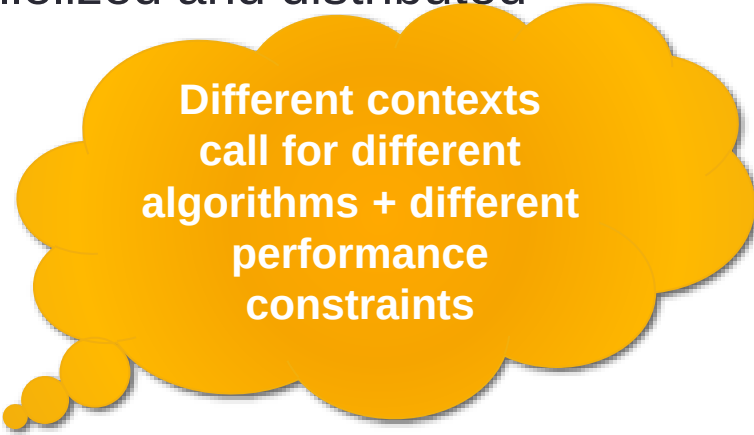
Do you need to keep track of any information?

Robot 1: can move North, East, West, South.

Robot 2: can move in four diagonal directions.

Computer = algorithm?

- Your experience: Java, Python, C, etc.
- Other computer models:
 - Quantum computers
 - DNA computation
 - Supercomputers / Datacenters with parallelized and distributed computation
 - Power-sensitive computation for mobile
 - Embedded circuits



**Different contexts
call for different
algorithms + different
performance
constraints**

What is a computer/computation/algorithm?



Bird's eye view of CSE 331

- Pick a model of computation
- Study what problems it can solve
- Prove its limits

Bird's eye view

- Pick a model of computation

Study what problems it can solve

- Prove its limits

Classification: is input of type A or not?

e.g. is n prime? is list sorted?

Computation: for specific input, what value should I output?

e.g. what's min cost of Hamiltonian tour?

Bird's eye view

- Pick a model of computation

Study what problems it can solve

- Prove its limits

Classification: is input of type A or not?

Decision problem

Computation: for specific input, what value should I output?

Function problem

Bird's eye view

- Pick a model of computation

Study what problems it can solve

- Prove its limits

Classification: is input of type A or not?

Decision problem

Computation: for specific input, what value should I output?

Function problem

Bird's eye view

- Pick a model of computation

Study what problems it can solve

- Prove its limits

Classification: is input of type A or not?

Decision problem

$\{ w \mid w \text{ is of type } A \}$

$\text{PRIME} = \{ 2, 3, 5, 7, \dots \}$

$\text{SORTED} = \{ \langle 1, 3 \rangle, \langle -1, 8, 17 \rangle \dots \}$

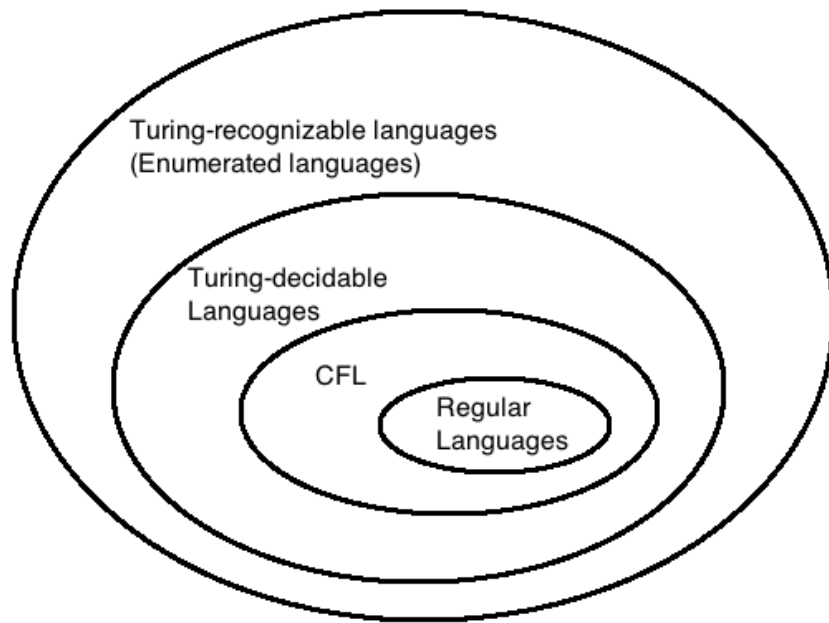
Decision problems are coded
by sets of strings

Models of computation

- Finite automata
- Context-free grammars
- Turing machines

Models of computation

- Finite automata
- Context-free grammars
- Turing machines



So let's get going 😊

- **Textbook** reference: Chapter 0, Section 1.1
- **First topic:** Automata (DFA)

Code input as strings
Model memory using states

- Natural language processing
- Hardware design
Moore machines
- Controllers / Robots
SPIS!

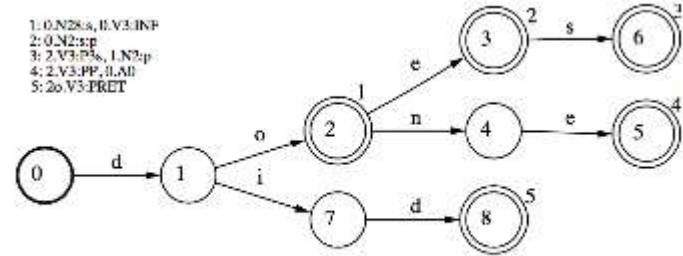
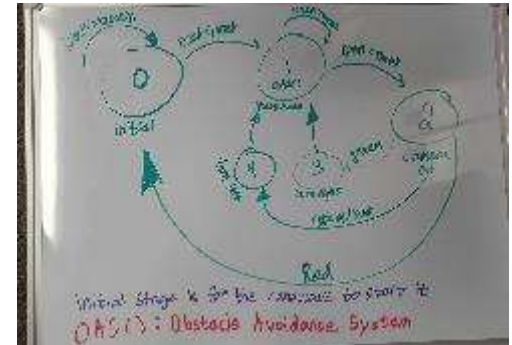


Fig. 1. Representation of dictionaries by automata.



Example: subway turnstile

- A subway turnstile is locked until a token is entered, at which point it unlocks in response to a single push, after which it locks again.

When you approach the turnstile, will it open?

How can we model this problem as a classification question?

Example: subway turnstile

- A subway turnstile is locked until a token is entered, at which point it unlocks in response to a single push, after which it locks again.

When you approach the turnstile, will it open?

How can we model this problem?

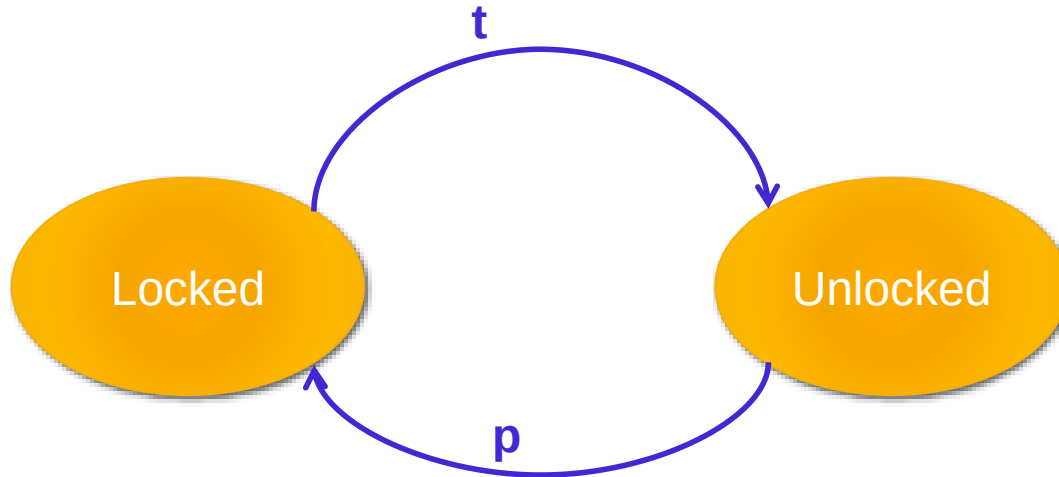
Alphabet: {token entered, **push**}

Input: sequence of moves

Language: {sequences of moves giving unlocked gate}

Example: subway turnstile

- A subway turnstile is locked until a token is entered, at which point it unlocks in response to a single push, after which it locks again.

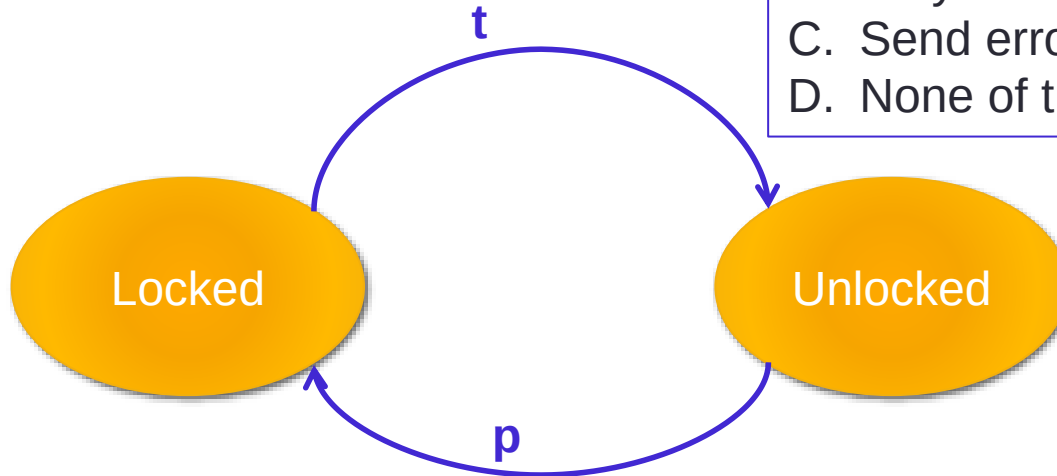


Example: subway turnstile

- A subway turnstile is locked until it is pushed, at which point it unlocks in response, and then locks again.

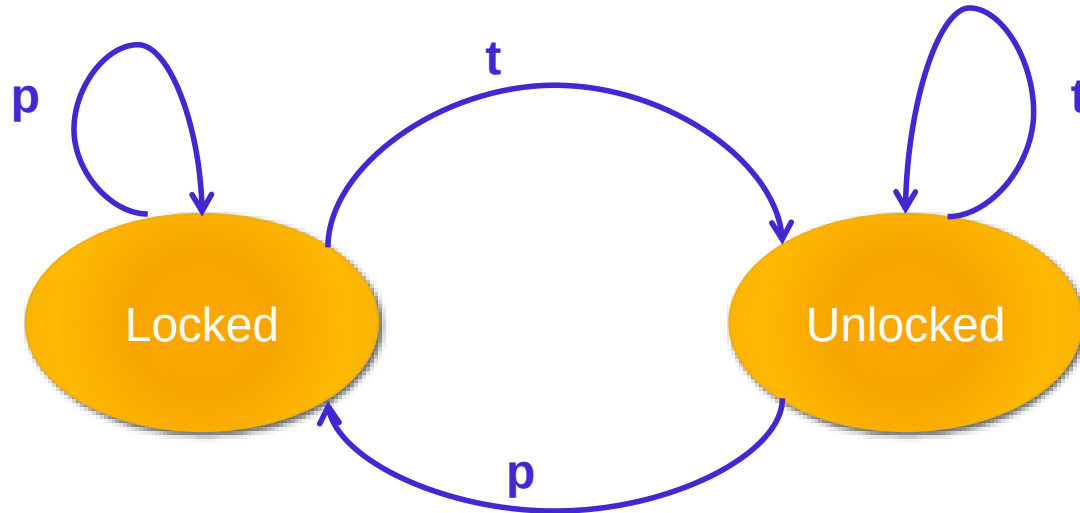
What happens if the turnstile is pushed while locked?

- A. Transition to unlocked
- B. Stay locked
- C. Send error message
- D. None of the above



Example: subway turnstile

- A subway turnstile is locked until a token is entered, at which point it unlocks in response to a single push, after which it locks again.

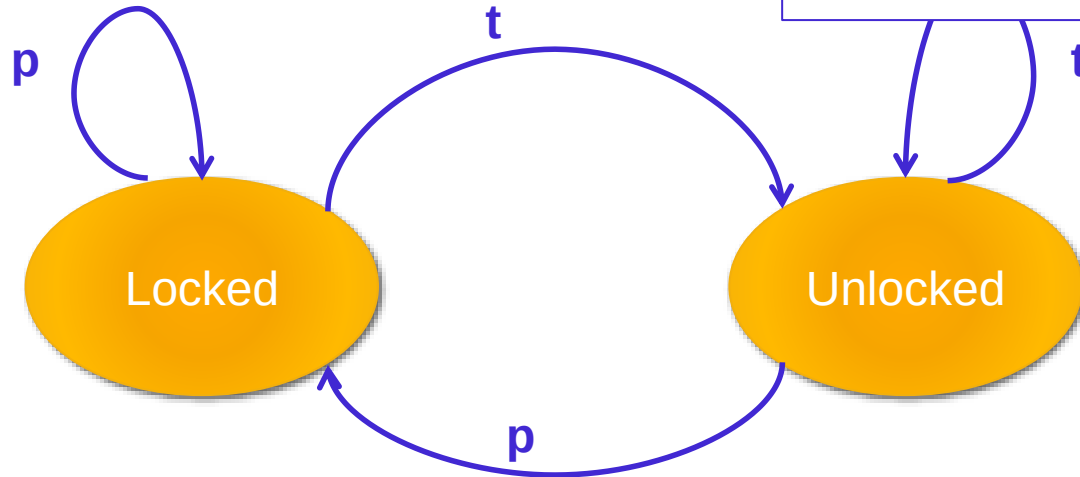


Example: subway turnstile

- A subway turnstile is locked until a coin is inserted, at which point it unlocks in response. When a person passes through, it locks again.

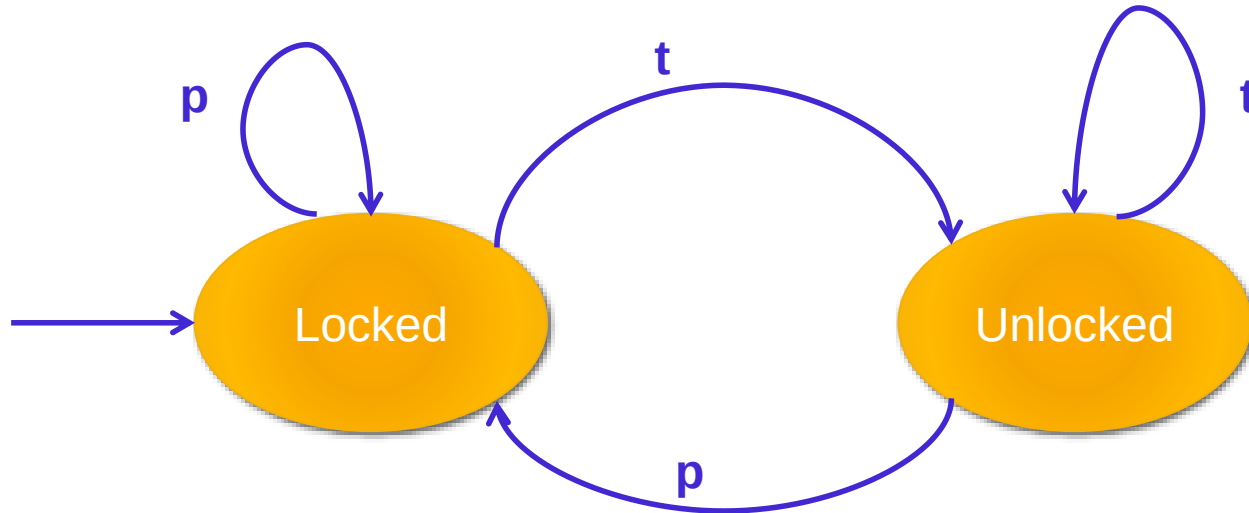
What's the initial state of the turnstile?

- A. Locked
- B. Unlocked



Example: subway turnstile

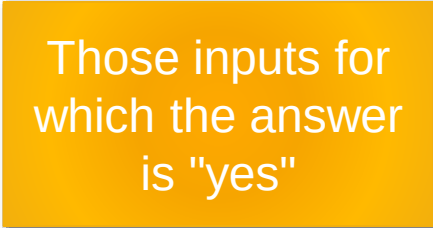
- A subway turnstile is locked until a token is entered, at which point it unlocks in response to a single push, after which it locks again.



Finite automata and decision problems

- Decision problem "is w of type A ?" coded by the set

$$\{ w \mid w \text{ is of type } A \}$$



Those inputs for
which the answer
is "yes"

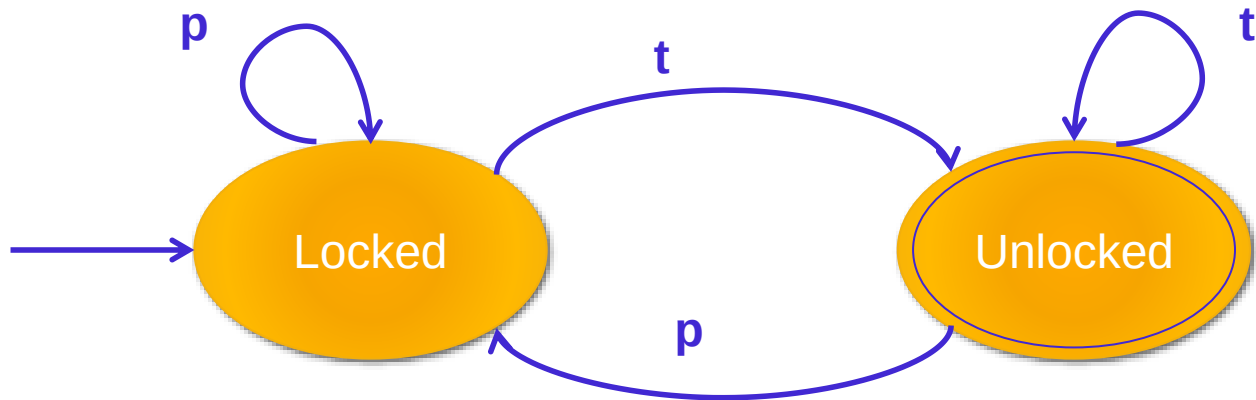
A problem is **solved** by a finite automaton if
the machine *accepts* all w for which the answer is yes and
rejects all w for which the answer is no

Example: subway turnstile

Problem: Which sequences of moves will unlock turnstile?

Alphabet: $\{t, p\}$

Language: $\{\text{sequences of moves giving unlocked gate}\} = \{t, ppt, \dots\}$
 $= \{\text{strings over alphabet that, when input to this automaton, end in Unlocked state}\}$



What is an alphabet?

An **alphabet**, denoted by Σ , is a finite, non-empty set of symbols.

- $\Sigma = \{a, b\}$
- Unary alphabet $\Sigma = \{1\}$
- Binary alphabet $\Sigma = \{0, 1\}$
- English alphabet $\Sigma = \{a, \dots, z, A, \dots, Z\}$
- Alphanumeric alphabet $\Sigma = \{a-z, A-Z, 0-9\}$

Powers of an alphabet

- Σ = Some alphabet
- Σ^k = Set of all strings of length k over Σ
- $\Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \dots =$ Set of all strings over Σ . Σ^* is the universal set of all strings
- $\Sigma^+ = \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \cup \dots =$ Set of nonempty strings over Σ

What is a string?

- A **string** or word is a finite sequence of symbols chosen from Σ .
- A string $x \in \Sigma^*$. An empty string is denoted by ε .
- $|x|$ = length of string x
- $n_\sigma(x)$ = #occurrences of symbol $\sigma \in \Sigma$ in the string x

Examples of string

- $x = \text{abaaabb}$ from $\Sigma = \{a, b\}$
- $x = 111$ from $\Sigma = \{0, 1\}$
- $x = \varepsilon$ from $\Sigma = \{a, \dots, z, A, \dots, Z\}$
- $x = \text{Bond007}$ from $\Sigma = \{a - z, A - Z, 0 - 9\}$

What is a language?

A **language** over Σ is a subset of Σ^*

- The empty language $\varnothing$.
- $\{\epsilon, a, aab\}$ - a finite language.
- Language of palindromes over $\{a, b\}$
- $\{x \in \{a, b\}^* \mid n_a(x) > n_b(x)\}$. $\{x \in \{a, b\}^* \mid |x| \geq 2 \text{ and } x \text{ begins and ends with } b\}$

Vocabulary review

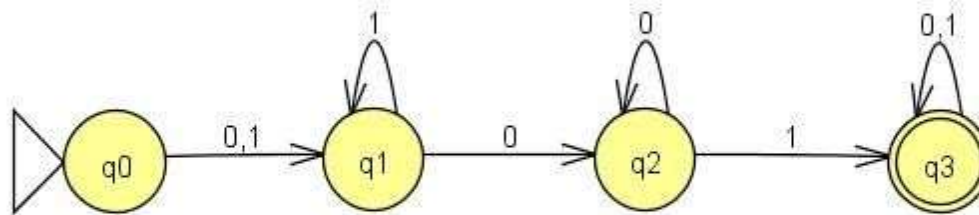
Sipser p. 14

- $\{a, b, c, d, e\}$ The set whose elements are a, b, c, d, e
- $\{a, b\}^*$ The set of finite strings over a, b
 - Includes empty string ϵ
 - Includes a, aa, aaa
 - Includes b, bb, bbb
 - Includes $ab, ababab, aaaaaaabbb$
 - Does **not** include infinite sequences of a 's and b 's
 - Has **infinitely many** different elements
- $|ababab| = 6$ The length of the string $ababab$ is 6
- $|\{a, b, c, d, e\}| = 5$ The size of the set $\{a, b, c, d, e\}$ is 5
- A **language** (over alphabet Σ) is a set of strings (over Σ)

Deterministic Finite automaton

Sipser p. 34, 40

- Input: finite **string** over a fixed **alphabet**
- Output: "accept" or "reject"



Start state
(triangle/arrow)

Accept state
(double circle)

The **language** recognized by the machine is the set of strings it accepts.

Deterministic Finite automaton

Sipser p. 34, 40

- **Computation** of the machine on an input string

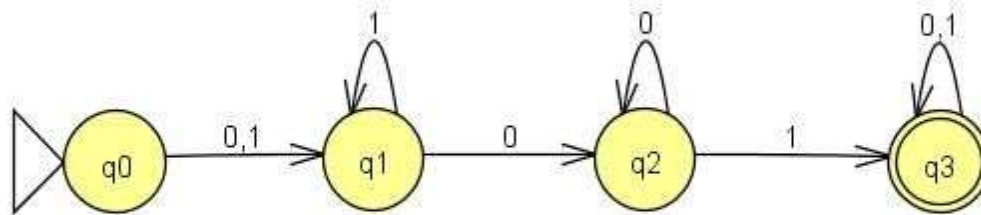
Sequence of states in the machine, starting with the initial state, determined by transitions of the machine as it reads additional input symbols.

How does the computation of the machine on a string relate to whether the string is accepted by the machine?

- A. If the **first state** in the computation is an accept state, the string is accepted.
- B. If **any state** in the computation is an accept state, the string is accepted.
- C. If the **last state** in the computation is an accept state, the string is accepted.
- D. If **all of the states** in the computation are accept states, the string is accepted.
- E. None of the above

Finite automaton

- Input: finite string over a fixed alphabet
- Output: "accept" or "reject"



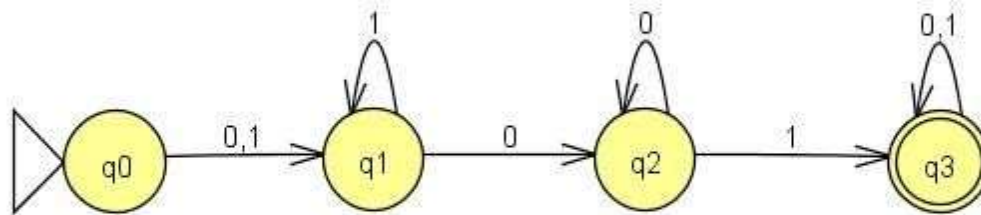
Start state
(triangle/arrow)

Accept state
(double circle)

Example input: 0001

Finite automaton

- Input: finite string over a fixed alphabet
- Output: "accept" or "reject"

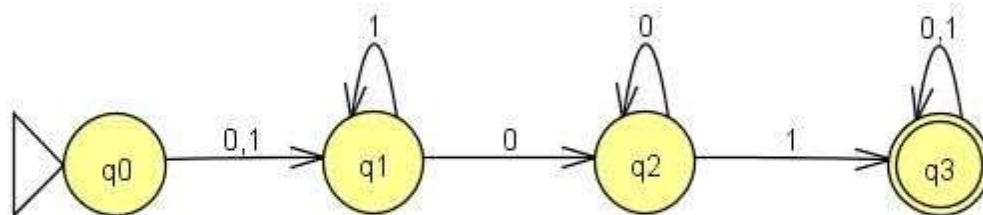


What is the computation of this machine on input **110**?

- | | |
|----------------------|-------------------|
| A. q1, q1, q0 | B. q1, q1, q2 |
| C. q0, q1, q2, q3 | D. q0, q1, q1, q2 |
| E. None of the above | |

Finite automaton

- Input: finite string over a fixed alphabet
- Output: "accept" or "reject"



Is the empty string accepted?

Is the string 001 accepted?

Is the string 0011010101010 accepted?

What is the language recognized by this machine?

For next time

- Set up course tools: Classroom
- Read all the questions + relevant examples in the book
- Read the book section (which section to read will be provided) before the class